First Program

→ getting started

Src code → Compiler → M'E format

PL

executable format

run

0/1

Binary language

a=1

b=2

c=a+b;

a bc

Compiler
↳ Translation
↳ find errors

IDE → Code Blocks → Youtube
VS Code

Replit → C++ file → run

Namaste Dunia

~~Hello World~~

flowchart :-

start

int main() {

}

Start of program

int main() {

}

→ Print "Namaste Dunia"

→ cout

cout

std

cout

→ << ? <<

cout << → to write

cout >> → compilation errors

→ endl → new line or enter

→ \n → enter

→ ; → to end line

Data types & Variables

int a = 5;

Data type

int a = 5

type variable

5

a

4 byte → 32 bit

→ Type of data

→ memory

→ char ch = 'a';

char ch = 'a';

type

'a'

ch

1 byte → 8 bit

a

→ bool → True, 1

False, 0

1 byte

bool b = 1;

1

b

→ float f = 1.2;

4 byte

1.2

f

→ double d = 1.23;

8 byte

1.23

d

Variable name

abc
ABC
A1
A_1

~~1abc~~ ✗
abc1 ✓
_a1 ✓

How data is stored?

int a = 8; → Binary → 1000 (4 bit)

int → 4 byte

8 8 8 8
00000 000 0000000 1000

32 bit

int a = 5 → 101 (3 bit)

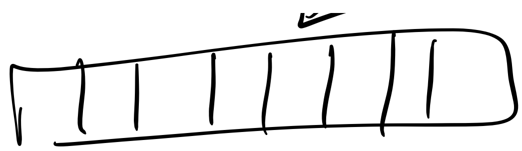
32 bit
00 0000000 101

-ve number → ?

→ char ch = 'a'; → ASCII table → H/W

char → 1 byte

'a' → 97 → Binary

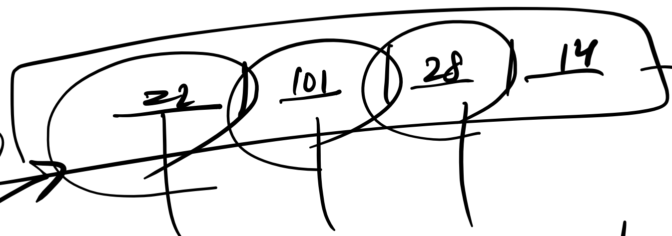


0/1

8 bit → cha

32 → int

→



1526

int a = 123;

a b c d

char / int

datatype sc

→ Type casting

a → 97

int a = 'a';

cout << a << endl; → ?

char ch = 98

Integer → 4 byte → 32 bit

max $2^{32}-1$

min = 0

2^{16}

char ch → 1 byte → 8 bit

max → $2^8 - 1$

min → 0

123456

int

64

64 → @

char

How -ve no are stored

First bit

↳ +ve → 0

↳ -ve → 1

-ve no → -5

display

↓ 2's compliment

↳ Ignore the -ve sign

5

↳ convert into binary rep.

0 00000000 101

↳ take 2's compliment & store

000 00000 0000 000 101

1's compl. → 11111111111 010

+1

2's comp 1 1111111111 011

-ve

int a = -5

print a = -1

-ve

1111 1111 1111 1111 1011

1's comp → 0000 0000 0000 0000 0100

+1

2's comp → 0000 0000 0000 0000 0101 → 5

print -5

compilation → Namaste Dunia

↳ Data types

↳ Variable

31

$2^{31}-1$

| 1 | 0 | |

| 1 | |

$-(2^{31}-1)$

$-(2^{31}-1), \ldots, 2^{31}-1$

$-2^{31}, \ldots, 2^{31}-1$

|0| ~~~ → 0.

|1| ~~~ → -0 → 0

1 no → "0" ← 2 representation

→ default

. int = 1 q -1

0 — — — $2^{32}-1$

unsigned int = 2

-112

+112

000 ... 1101

2's 0 1 1 1 1 1 1 1 1 | 0 101

print a unsigned

huge value

Operators :-

% → modulo op

Arithmetic

↳ +, -, *, /

2/5 → 0.4

→ 0 = ?

int/int → int

0.4 → 0

float/int = float

double/int = double

$\frac{2.0}{5} = 0.4$

int a = 0.4

a = 0

→ Relational Operators

==
>
<
>=
<=
!=

a=3 a equal to b?
b=4

a==b

a>b
a<b
a>=b
a<=b
a!=b

int a = 3;

→ Logical op → &&
||
!

23 && 6foot && 80kg → fit

23 || 6foot || 80kg → fit / None

a=23

!a → 0

→ Bitwise operators

Compilation

First progr → Namaste Duniya

Line

Data types / Variable

+ve / -ve / unsigned / signed

Op → Arith / Rel / Logical

Con →

Ce

Hw → Typecasting

ck cheez

Revise

Programming Language